MA1905 – Unit 2 Practice

1. Translate each phrase into an algebraic expression. Use x as the variable.

a) Nine less than twice a number.

b) Three times the difference between a number and 7.

c) The quotient of a number and 5, increased by 4.

d) Two-thirds of the sum of a number and 12.

2. Convert each expression to computer notation.

a) $8x^3 - 5y$

b) $\frac{2x-7}{x+4}$

c) $\sqrt{5x-1}$

d) $x^{\frac{5}{2}} + 3xy$

3. Convert each expression from computer notation to standard written mathematical notation.

a) 6 * x^3 + 2 * x

b) (4 * y - 1) / (y + 6)

c) sqrt(x + 9) - 2 * y^2

4. Write a computer assignment statement for each situation.

a) Store the product of length and width in a variable called are

b) Calculate the total cost of n tickets at $18.50 each and store the result in total.

c) Calculate the average of test1, test2, test3, and test4 and store the result in average.

d) Store the text "Pending" in a variable called status.

5. Translate each statement into an inequality. Use x as the variable.

a) A package may weigh at most 23 kg.

b) A mark below 60% requires reassessment.

c) A user must be at least 18 years old.

d) A temperature greater than 85°C causes an alert.

6. Set up and solve an appropriate equation for each of the following. Clearly indicate what the variable represents. Show all steps and state your final answer.

a) A company ordered 42 computer monitors and keyboards altogether. The company ordered 8 more monitors than keyboards. How many monitors and how many keyboards were ordered?

b) A student purchased a computer accessory. After 15% sales tax was added, the total cost was $172.50. What was the price of the accessory before tax?

c) Three consecutive whole numbers have a sum of 204. Find the three numbers.

d) A rectangular room is 5 m longer than it is wide. Its perimeter is 50 m. Find its dimensions.

7. Set up and solve an appropriate inequality for each of the following. Clearly indicate what the variable represents. Show all steps and state your final answer.

a) A student has already saved $275 toward a new computer. The student saves $45 each week and wants to have at least $680. How many weeks must the student continue saving?

b) A department has $730 available for monitors. It has already spent $164 on cables and adapters. The cost of each monitor is $84. What is the maximum number of monitors that can be purchased?

c) A computer lab has 240 printer cartridges in storage. Each month, 15 cartridges are used. For how many months will there be more than 90 cartridges remaining?

8. Trace the assignment statements. Write the value stored in each variable after each line is executed.

Statement	Values after execution
x = 5	
y = 3	
z = x * y	
x = z + 2	
y = x - 4	
z = y + x	